

Rough jump modes

Driver jumps vs direct jumps: a comparison for your decision. Session H, 15 September 2026.

The choice

Session F put Hawkes jumps on the lifted rough Heston host in two ways. Both are implemented in `models/jump_hosts.py`, and the default is still driver mode.

	driver (current default)	direct
where the jump enters	the Volterra driver, like dt and dW : $dU_i = \dots + J dN$	a separate factor: $V = v_0 + w^T U + D$, $dD = -\lambda_D D dt + J dN$
V 's response to a jump at s	$J \cdot R(t - s)$, the kernel's resolvent: $J t^{(\alpha-1)} E_{\alpha, \alpha}(-\kappa t^\alpha)$	$J e^{(-\lambda_D (t - s))}$, plus the factors' reaction to the raised V
extra parameter	none (the decay is the kernel's, so H)	λ_D (default κ)
still affine?	yes: the same lifted Riccati, jump transform added to $F(\psi)$	yes: one more factor

Everything below was measured with `study_jump_modes.py`: $\kappa = 3$, $\theta = 0.04$, $\xi = 0.3$, $H = 0.12$. The results are in `captures/jump_modes.json` and the figure is at the end.

What was measured

1. HOW A JUMP DECAYS

These are responses to one jump with no excitation. Each is scaled to add the same variance over its first trading day.

mode	share of the 30-day impact in the first hour / day / week	V at 5 min ÷ V at 1 day	V at 1 week ÷ 1 day	V at 30 days ÷ 1 day
driver, shipped lift (N = 24)	6.1% / 19.0% / 45.9%	8.3	0.42	0.10
driver, finer lift (N = 40)	6.3% / 19.1% / 46.0%	9.8	0.42	0.10
direct, $\lambda_D = \kappa =$ 3/y	1.0% / 6.0% / 26.3%	1.1	0.80	0.33
direct, $\lambda_D = 25/y$ (10-day e-fold)	2.9% / 17.0% / 62.7%	1.2	0.53	-0.05
Heston host (reference)	0.6% / 3.9% / 19.2%	1.0	0.95	0.71

Driver mode has the shape of an aftershock sequence. It spikes hard within the hour, falls as a power law, and still carries a tenth of its one-day level after a month. Power-law relaxation of volatility after crashes (an Omori law) is a documented market regularity (Lillo & Mantegna, 2003).

Direct mode is an exponential, so λ_D alone sets its horizon. At the default $\lambda_D = \kappa$ it barely decays inside a month. At 25/y the response **undershoots**, and V ends up below its no-jump path after about 20 days. The mechanism: while D is high, mean reversion pushes the rough factors down. Those factors remember through the kernel after D has gone. This is a side effect of wiring D into V's drift, not a property anyone would choose.

2. WHAT A DRIVER JUMP'S "SIZE" MEANS

In the exact rough model the kernel is infinite at zero, so a driver jump has no instantaneous size in V. It only has an integrated impact. On a lift the instantaneous move is finite, but it depends on the time step and the lift:

driver jump, per unit J	step 5 min	step 1 h	step ¼ day
V moves at once, shipped lift	40.2	17.8	15.0
V moves at once, finer lift	47.4	18.5	15.4
1-day integrated impact, shipped (exact 0.03358)	0.03305	0.03178	0.03346
1-day integrated impact, finer	0.03340	0.03213	0.03381

A direct jump moves V by exactly J at any step. **Consequence:** in driver mode, J must be specified, calibrated and reported as integrated variance impact. A statement such as “V jumps by 0.01” means something different on every grid. The integrated impact is stable to within 5% across steps and lifts.

3. THE SECOND-SPIKE PROPERTY

The question is whether the variance increment after the second of two clustered shocks is larger than after the first. The test uses the exact superposition identity from Session F on the same 21 (gap, window) pairs, with quarter-day steps.

mode	branching ratio 0.6	branching ratio 0.95	gaps where it holds
driver, shipped lift	0 of 21	14 of 21	½ to 10 days (at 0.95)
driver, finer lift	0 of 21	14 of 21	identical
direct, $\lambda_D = 3/y$	6 of 21	19 of 21	½ to 2 days at 0.6
direct, $\lambda_D = 25/y$	5 of 21	10 of 21	½ to 2 days at 0.6

In driver mode the kernel’s early decay beats the excitation unless clustering is near-critical. Branching ratios near one have been reported for index futures at high frequency (Hardiman, Bercot & Bouchaud, 2013), so this regime is not exotic. Direct mode delivers the property at moderate clustering whenever the excitation outruns λ_D . The lift choice does not affect it.

4. KURTOSIS (SESSION F)

Hawkes daily returns are more kurtotic than Poisson in both modes: driver 6.73 ± 0.17 vs 4.43 ± 0.08 , direct 8.35 ± 0.21 vs 5.28 ± 0.09 , a ratio of about 1.5 in each. The two modes' jump sizes were not matched in integrated impact there, so the levels are not comparable across modes. Kurtosis does not separate them.

5. CAN THE DATA DECIDE?

The planned test runs an event study on the recorded history. It takes daily realised variance after large-return days and asks whether the excess decays as a power law or exponentially. Its power was measured on simulated 3-year histories with realistic settings: Hawkes $n = 0.6$, about 10 events per history, and jump sizes matched in 1-day impact.

truth	right shape wins, single 3-year history	pooled over 36 years
driver	9 of 12	power law ($p = 0.59$) ✓
direct ($\lambda_D = 25/y$)	4 of 12	exponential (0.082/day) ✓

One 3-year history cannot decide this. The test leans toward “power law”, which it reports 8 times out of 12 even when the truth is exponential. A power-law verdict on SPY would carry a likelihood ratio of about 1.1, which is not evidence. Both modes were simulated from the same random draws, and their verdicts agree on 11 of the 12 histories. What decides a verdict is the draw, not the mode.

Two routes would decide it:

- **A decade or more of realised variance.** 36 simulated years separated the modes. This needs longer history than the recorder's default pull.
- **The option surface after a scheduled shock.** Driver mode says the extra variance priced to expiry T after an event grows like $(T - t_e)^\alpha$, which is concave with no ceiling. Direct mode says it saturates at J/λ_D . Chains recorded before and after an FOMC day measure exactly that curve, and the 16 September decision falls inside this week's recording window. The clock estimator (`calibrate/clock.py`) already fits the event step; extending it with a post-event response term is a small change.

Recommendation

Keep driver mode as the default, with three changes and one open check:

1. **Parameterise driver jumps by integrated impact:** variance added over the first day, or over the kernel's own horizon. Do not use a V jump size, which depends on the grid (section 2).
2. **Do not claim the second-spike property at moderate clustering on the rough host.** State its actual condition: branching ratio near one. If the property is a hard requirement at $n \approx 0.6$, only direct mode delivers it (section 3).
3. **If direct mode is ever used, fix λ_D from data and disconnect D from the factors' mean reversion**, or accept the undershoot (section 1).
4. **Run the option-surface test on the September 16 FOMC chains** once recorded. It is the most direct measurement available this month that can tell the two modes apart (section 5).

Why driver mode: it is the Volterra-consistent choice, and one H then governs both diffusive roughness and jump relaxation. It adds no parameter, it keeps pricing in the same Riccati, and its aftershock shape matches the empirical Omori-law literature. Direct mode is two models glued together, with a decay rate that the rough part does not know about.

Your call: confirm driver as the default, or name the property (second spike at moderate clustering) that would make direct mode the requirement.

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